

COLX 585: Membership Attack in VLMs Project: Milestone 5

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Abstract

This paper addresses the problem of membership inference attack (MIA) against pre-trained large vision-language models. Three MIA methods are explored, implemented, and evaluated by improving the previous baseline model from milestone 4. In the first method, temperature-sensitivity features are added to the existing model to categorize membership by querying the same set of data at different temperatures. The second method builds upon the temperature-sensitivity model and adds MaxRenyi-K% and ModRenyi features. Lastly, the third method added image-based features to verify a signal through images memorized by the model potentially producing embeddings more immune to corruption than unseen images.

1 Introduction

Membership inference attack (MIA) is an important privacy concern for large vision-language models, since it aims to determine whether a given image-text sample was part of a model's training data. In realistic settings, this is challenging because attackers often do not have access to model parameters or token-level logits, so useful membership signals must be inferred from model outputs. In this milestone, we build on our milestone 4 baseline and evaluate three feature-based attack variants in a modular pipeline. We first add temperature-sensitivity features, and then extend the model further with MaxRenyi-K% and ModRenyi features. Lastly, we add image corruption features. Our results show modest improvements over the baseline, with temperature-based features providing the clearest gains, while the Renyi-based and image-based features appear to add only limited additional signal.

2 Background

Summary of related research papers:

2.1 Paper 1: "Samira, D., Habler, E., Elovici, Y., & Shabtai, A. (2025). Variance-based membership inference attacks against large-scale image captioning models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 9210-9219)."

The paper addresses the problem of membership inference in large-scale image captioning models. More specifically, it asks whether an attacker can determine if a given image was part of a model's training set, under a realistic black-box setting where the attacker can only query the model with images and observe the generated captions, without access to model parameters or ground-truth training captions. The evaluation is conducted on a composite benchmark built from several public image-captioning datasets: MSCOCO, Flickr30k, TextCaps, IAPR TC-12, and NoCaps. In addition, the authors construct an artificial non-member dataset using Stable Diffusion v1.5, generating images from Flickr30k captions in order to simulate images that were not seen during training. The paper's setup treats one dataset as the target model's training source and samples non-members from the remaining datasets.

The target models are three publicly available large-scale image captioning systems from Hugging Face: BLIP Large, ViT-GPT2, and GIT Base. Beyond these target models, the attack pipeline also relies on a pre-trained text encoder to embed generated captions and a pre-trained image encoder to represent images for downstream attack classification.

The paper makes four main contributions. First, it proposes two new attacks for image captioning models, MVTA and C-WSA, designed for zero-knowledge and partial-knowledge settings, respectively. Second, it introduces a new means-of-variance (MV) metric to capture the dispersion

of caption embeddings generated from the same image. Third, it improves weakly supervised attack training through a confidence-based pseudo-member filtering mechanism. Finally, it validates these methods on realistic public models rather than toy or heavily simulated settings, showing that the proposed attacks outperform prior baselines and expose a meaningful privacy risk in operational image captioning systems

2.2 Paper 2: "Yihao, L. I. U., & Xinqi, L. Y. U. LOMIA: Label-Only Membership Inference Attacks against Pre-trained Large Vision-Language Models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*."

This paper studies membership inference against pre-trained large vision-language models under a label-only setting. More specifically, it asks whether an adversary can infer if a given image-text sample was used during pre-training when the target model exposes only its final output, rather than token-level logits or internal parameters. The problem is framed as a more realistic and more restrictive privacy auditing setting for deployed VLLMs.

The authors construct two multimodal evaluation sets for the attack. The first, LOMIA/LAION, is built from samples drawn from the overlap of the pre-training corpora used by the target VLLMs, including LAION, Conceptual Captions 3M, Conceptual 12M, and SBU Captions. The second, LOMIA/CC, is built from MS COCO. In both cases, member samples are taken from data known to be used in pre-training, while non-member samples are synthesized with Stable Diffusion v1.5 from corresponding captions. Each dataset contains 600 images in total, with 300 members and 300 non-members.

The main target models are three open-source pre-trained VLLMs: LLaVA-1.5, MiniGPT-4, and LLaMA-Adapter V2.1. The paper also reports results on the closed-source model GPT-4o. For the attack pipeline itself, the authors use LLaVA-1.5-7B as the surrogate model, all-MiniLM-L6-v2 for text-text similarity, and CLIP for image-text similarity.

The paper's main contribution is the introduction of LOMIA, a label-only membership inference framework tailored to pre-trained VLLMs. Within this framework, the authors propose three attack variants—TTFA, ITFA, and DUFA—which

use semantic similarity features to approximate generation perplexity and then use that signal for membership prediction. Conceptually, the work is important because it shows that effective MIAs are still feasible even when logits are unavailable, and empirically it demonstrates that these label-only attacks are competitive with, and in some settings comparable to, stronger logits-based baselines.

3 Milestone 5

For this milestone, we continue with the more modular structure of using multiple .py scripts in conjunction with a slim notebook to interface with colab. Our code is in the following files:

batches.py: handles memory-efficient feature extraction over large datasets by processing samples in chunks, calling the feature extraction functions from `features.py` on each batch, and returning the results as a combined DataFrame. This file includes the following functions:

- **analyze_sample_batch:** Computes features for a list of (image, text, label, id) tuples, optionally using a base model for comparative features. Returns a DataFrame.
- **process_dataset_in_batches:** Chunks a DataFrame and calls `analyze_sample_batch` on each chunk, clearing CUDA cache between batches. Returns a concatenated DataFrame of all results.

classifier.py: contains the implementation of classifiers and their associated helper functions. The function predicting the labels and generating the submission file for the test data set is also contained here. This script includes the following functions:

- **_clean:** Replaces infinities and NaNs with median values for preprocessing.
- **_get_feature_cols:** Auto-detects numeric feature columns, excluding metadata and constant columns.
- **build_membership_classifier:** Trains Logistic Regression, XGBoost, and an ensemble on extracted features; returns classifiers, scalers, and validation metrics (AUC, TPR@FPR=0.1).

- **_predict:** Generates membership probability predictions from a classifier result dict, handling ensemble averaging.
- **generate_submission:** Creates and saves a Kaggle-format submission CSV with predicted membership probabilities.

config.py: The python file where global variables are defined.

features.py: The python file containing functions to create features, including helper functions. It contains:

- **_load_image:** Loads an image from raw bytes or PIL Image and converts to RGB.
- **_prepare_inputs:** Splits text into user/assistant parts, applies chat template, and tokenizes inputs with prompt length info.
- **_compute_loss_features:** Computes features, including caption loss, perplexity, token probabilities, entropy, and temperature sensitivity at multiple temperatures.
- **compute_vlm_features:** Runs the fine-tuned model and adds zlib compression ratio features to the core features.
- **compute_reference_features:** Runs the same feature extraction on the base (pre-trained) model, prefixing results with `ref_`.
- **compute_comparative_features:** Computes difference and ratio features between fine-tuned and base model outputs — the primary MIA signals.

milestone5.ipynb: Slim notebook that controls the model’s pipeline.

models.py: Python code that handles loading the fine-tuned and base SmolVLM models, processors, and dataset splits needed for MIA feature extraction. This script includes the following functions:

- **get_model_dtype:** Returns `bfloat16` or `float32` based on device capability.
- **_load_vlm:** Loads a SmolVLM model and processor from HuggingFace with appropriate dtype/attention settings.
- **load_model_and_processor:** Loads the fine-tuned target VLM.

- **load_base_model_and_processor:** Loads the base pre-trained SmolVLM for reference comparison.
- **load_data:** Loads the HuggingFace dataset and returns train/validation/test splits as DataFrames.

3.1 Method 1: Temperature-based Attack

Using one of the techniques described in "Membership Inference Attacks Against Vision-Language Models." by Hu et al, we added temperature-sensitivity features to our baseline model. The target-only inference technique described in the paper categorizes membership by querying the same set of data at different temperatures. Since training data members are more sensitive to changes in temperature, the model’s outputs should change significantly for member data after temperature shifts. The intuition is that data that were included in the training set produce more peaked token probability distributions, so their loss drops more sharply at low temperatures compared to non-members. (Hu et al. 2025)

The new features created for this model are:

loss_at_T025: Caption loss computed with temperature 0.25.

loss_at_T05: Caption loss at temperature 0.5.

loss_at_T20: Caption loss at temperature 2.0.

loss_at_T40: Caption loss at temperature 4.0.

temp_sensitivity: Difference between `loss_at_T05` and `loss_at_T20` — captures how much loss changes across a moderate temperature range. More negative = more peaked distribution = likely member.

temp_sensitivity_ext: Difference between `loss_at_T025` and `loss_at_T40` — same idea but across a wider temperature range for a stronger signal.

While this model slightly increased our score from the XGBoost baseline of $AUC=0.7793$, it is a small increase. The ensemble of the models that include temperature-based attack features has $AUC= 0.7846$.

Table 1: Performance Comparison of Classifiers and Baselines

Method	AUC	TPR@FPR=0.1
Logistic Regression	0.7841	0.2700
XGBoost	0.7816	0.2833
Ensemble	0.7846	0.2767

Table 2: Top 10 Individual Feature AUCs for Temperature-based Attack Model

Feature Name	Individual AUC
caption_loss_diff	0.7299
perplexity_ratio	0.7299
temp_sensitivity_ext_diff	0.7005
temp_sensitivity_diff	0.6906
ref_min_k10_prob	0.6879
ref_loss_at_T05	0.6864
ref_caption_loss	0.6813
ref_caption_perplexity	0.6813
ref_temp_sensitivity	0.6806
ref_min_k5_prob	0.6801

This model scored AUC=0.77825 in its Kaggle submission.

3.2 Method 2: MaxRenyi-K% and ModRenyi

In method 2, we expand on the pipeline in method 1, from which there was improvement on the baseline model, by using two techniques described in “Membership Inference Attacks against Large Vision-Language Models” by Li et al: MaxRenyi-K% and ModRenyi. Researchers in this paper had success with their two proposed MIA metrics, achieving an AUC of 0.815 on GPT-4 in multi-modal (i.e., image and text) MIA. Li et al describe MaxRenyi-K% as “utilizing the Renyi entropy of the next-token probability distribution on each image or text token” (Li et al. 2024). They use this metric because they posit that if a model has seen the image or text before, it will be more confident in the next token and therefore have a smaller Renyi entropy. While this metric is similar to the Shannon entropy and Min-K% probabilities that we have incorporated into our baseline model pipeline, MaxRenyi-K% combines both concepts by using Renyi entropy with richer distributional information (i.e., higher sensitivity to larger probabilities by specifying alpha, which adjusts the weights of probabilities in the distribution) applied to the most uncertain token positions. ModRenyi is an extension of this metric: the mean Renyi entropy across all caption tokens with no K% selection. This acts as a standalone metric, used as a single score per sample and capturing a global average distributional shape rather than specific token positions. Therefore, the following features are added in this method:

MaxRenyi-K% at $k = 0.10$ or $k = 0.20$ and alpha

$= 0.5$ or $\alpha = 2.0$

ModRenyi at $\alpha = 0.5$ or $\alpha = 2.0$

Alpha = 0.5 was chosen for giving more weight to the tail of the probability distribution (smaller probabilities), while $\alpha = 2.0$ was chosen for giving more weight to higher probabilities.

In addition to the logistic regression, XGBoost, and ensemble classifiers used in the baseline, we tried random forest, gradient boosting, and MLP (multilayer perceptron) to see which classifier would perform best. The former two models are extensions of our existing methods, using decision trees that capture nonlinear relationships, but with different implementations. Our goal was to test how these models handle feature relationships. The latter model was chosen for its ability to learn nonlinear transformations of our features, and potentially perform better on partially-seen non-members. However, our results show that the ensemble method outperformed these classifiers, with AUC = 0.7879 on the validation set.

Notably, the top 10 features with the highest AUC scores remain very similar to those of method 1. While there is a small improvement in the final AUC score by adding the MaxRenyi-K% and ModRenyi features, it is evident that they are redundant with existing features, as they do not provide strong signals of membership. Our boost in performance is likely due to the noise these features add to the classification. However, this is still informative, as it shows us that Renyi entropy does not provide much additional signal beyond Shannon entropy and Min-K%, and highlights what features are more important.

Table 3: Method 2: Model Performance Comparison

Method	AUC	TPR@FPR=0.1
Ensemble	0.7879	0.2867
Logistic Regression	0.7864	0.2967
Gradient Boosting	0.7848	0.3000

This model scored AUC=0.78038 on the Kaggle test set.

3.3 Method 3: ICIMIA - inspired image corruption

With the introduction of ICIMIA-based image corruption features, we intended to explore the methods put forward in the Wu et al., 2026 paper and verify if separate text-based and image-based features could provide a combined signal to the model

Table 4: Method 2: Top 10 Individual Feature AUCs

Feature Name	Individual AUC
caption_loss_diff	0.7301
perplexity_ratio	0.7301
temp_sensitivity_ext_diff	0.7009
temp_sensitivity_diff	0.6904
ref_min_k10_prob	0.6872
ref_loss_at_T05	0.6863
ref_caption_loss	0.6812
ref_caption_perplexity	0.6812
ref_temp_sensitivity	0.6804
loss_ratio	0.6796

for VLM-MIA. This method proposes that images seen during training are memorized by the model, such that when a small corruption is applied, the model is still able to recognize the image and will subsequently produce a similar embedding for the corrupted image as to the original. In contrast, an image not yet seen by the model, when corrupted, results in a more noticeably different embedding to the original unseen image embedding. Cosine similarity is used to compare original and corrupted image embeddings, for seen and unseen images. Member images are distinguished from non-member images based on a higher cosine similarity score, since the original and corrupted image embeddings remain closer together.

An advantage of this approach is that it does not require a reference model, nor any text processing.

We adapted the ICIMIA paper methods due to time and resource constraints. Instead of applying Gaussian blur, Motion blur and JPEG compression, we applied Gaussian blur only for image corruption. We used Gaussian blur at a mild level (radius = 2) and moderate level (radius = 5).

Features introduced:

img_sim_blur_r2 : mean patch cosine similarity between the blurred and original image at radius = 2

img_sim_blur_r2_min : minimum patch cosine similarities at radius = 2

img_sim_blur_r2_std : standard deviation of patch cosine similarities at radius = 2

img_sim_blur_r5 : mean patch cosine similarity between the blurred and original image at radius = 5

img_sim_blur_r5_min : minimum patch cosine similarities at radius = 5

img_sim_blur_r5_std : standard deviation of patch cosine similarities at radius = 5

The conjecture that the ICIMIA features combined with existing text-based features might together produce a signal did not meaningfully improve model performance. Despite these image-based corruption features being a departure from existing features in the model, they did not add any substantive signal. The top 10 features using this method are almost identical to those in methods 1 and 2. This suggests that future work should incorporate features that address both image and text together. An example is the PC-MMIA approach (Zheng et al., 2025), which treats both modalities together, jointly capturing image and text memorization.

For overall pipeline efficiency, the function `compute_icimia_features()` allows cached embeddings to be passed in order to avoid repeated computation. We also changed `torch.no_grad()` to `torch.inference_mode()` in several pipeline functions.

Table 5: Method 3: Model Performance Comparison

Method	AUC	TPR@FPR=0.1
Logistic Regression	0.7804	0.2467
XGBoost	0.7798	0.2700
Ensemble	0.7819	0.2533

Table 6: Method 3: Top 10 Individual Feature AUCs

Feature Name	Individual AUC
caption_loss_diff	AUC=0.7299
perplexity_ratio	AUC=0.7299
temp_sensitivity_ext_diff	AUC=0.7007
temp_sensitivity_diff	AUC=0.6906
ref_min_k10_prob	AUC=0.6879
ref_loss_at_T05	AUC=0.6864
ref_caption_loss	AUC=0.6813
ref_caption_perplexity	AUC=0.6813
ref_temp_sensitivity	AUC=0.6806
loss_ratio	AUC=0.6793

This model scored AUC = 0.77626 on the Kaggle test dataset.

References

- D. Samira, E. Habler, Y. Elovici, and A. Shabtai. 2025. Variance-based membership inference attacks against large-scale image captioning models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9210–9219.
- Zhan Li, Yongtao Wu, Yihang Chen, Francesco Tonin, Elias Abad Rocamora, and Volkan Cevhar. 2024. Membership Inference Attack against Large Vision-Language Models. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.
- Yihao Liu and Xinqi Lyu. 2025. LOMIA: Label-only membership inference attacks against pre-trained large vision-language models. In *Proceedings of the 39th Annual Conference on Neural Information Processing Systems (NeurIPS)*.
- Zongyu Wu, Minhua Lin, Zhiwei Zhang, Fali Wang, Xianren Zhang, Xiang Zhang, and Suhan Wang. 2026. Image Corruption-Inspired Membership Inference Attacks against Large Vision-Language Models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026)*, pages 7945–7957.

4 Contributions

4.1 Milestone 5

Christina: Completed and reported model method 2. 0.25

Jennifer: Completed and reported model method 1. 0.25

Rachelle: Completed and reported model method 3. 0.25

Yucai: Literature review, 2 research papers. 0.25